Evaluation

Method

We ran a usability test to see how well people could actually use the AI Failures Museum. We focused on four main areas: whether the navigation was easy to find, if the layout worked on different screen sizes, if the design looked consistent across pages and whether the interface was engaging enough to keep people interested.We asked 20 people to use the system and fill out a quick survey with yes/no questions. This made it straightforward to spot patterns in what was working and what wasn't.

Participants

20 people took part in the evaluation on 15th March 2026. Everyone used their own devices and browsers, so we got a realistic picture of how the site performs in the real world. We kept it anonymous and didn't collect any personal information beyond the survey responses which was partly for privacy reasons, but also to keep things simple and in line with our ethics guidelines. Participation was completely voluntary.

Results

The results weren't great. All four areas showed significant problems:

1. Visual Clarity: Navigation Menu Usability

Only 30% (6 out of 20) found the navigation menu easy to locate and use. That means 70% struggled with it. Given that navigation is how people access the exhibits, a 70% failure rate is pretty bad as if users can't find the menu, they can't explore the museum at all.

2. Responsiveness: Browser Resize Behaviour

Again, only 30% said the layout adapted properly when they resized their browser window. 70% experienced some kind of layout problem. This is a major issue for a website that needs to work on phones, tablets, and desktops.

3. Consistency: Visual Design Coherence

40% found the fonts and button colours consistent across pages, while 60% noticed inconsistencies. This makes sense as we don't have a proper design system or style guide, so things look different depending on which page you're on. It's unprofessional and probably makes users wonder if they can trust the content.

4. Interface Engagement

This was the worst result: 0% found the interface engaging. Everyone (100%) said it wasn't captivating. While we prioritised making sure everything worked correctly over making it look pretty, an educational site still needs to hold people's attention. The interface clearly lacks visual interest as it doesn’t have any compelling imagery, no interactive elements, nothing to draw people in.

Summary of Key Findings

No metric achieved our target threshold of 70% positive responses. The evaluation identified four priority areas for improvement:

- Priority 1 (Critical): Navigation usability (70% failure rate)
- Priority 2 (Critical): Responsive design (70% failure rate)
- Priority 3 (Critical): Interface engagement (100% negative response)
- Priority 4 (Medium): Visual consistency (60% reported issues)

The core functionality works fine as people can browse exhibits, take quizzes and see their scores. But the user experience is rough enough that we'd need major improvements before rolling this out properly.

Evaluation of Other Success Measures

Quiz Comprehension: The system has recorded 3 quiz attempts from 2 users with an average score of 77.8% (attempts: 66.7%, 100%, 66.7%). This sample size is insufficient to draw conclusions about quiz difficulty or educational effectiveness.

Technical Reliability: Our automated test suite of 28 tests continues to pass at 100%, demonstrating core functionality (authentication, quiz scoring, comment posting) works correctly and reliably.

Limitations and Threats to Validity

Our evaluation has several important limitations that must be acknowledged:

Sample size: 20 participants is a decent number of participants, but our sample was entirely university students. The museum is supposed to be for students, professionals, and researchers, so we're probably missing perspectives from older or more experienced users.

No device control: We didn't track what devices or browsers people used. The 70% responsive design failure might look worse if lots of people tested on unusual screen sizes, or better if everyone just used desktop browsers.

Perception vs. reality: We asked people what they thought about the interface rather than giving them specific tasks to complete. A better approach would've been something like "Find an exhibit about healthcare AI failures and score above 70% on the quiz" and measure whether they actually succeeded. People's opinions don't always match their actual performance.

No comparison: We didn't test against another design or compare to similar websites. Is 30% success rate terrible for an early prototype, or does it mean we've got fundamental problems? We don't really know.

No follow-up questions: Binary yes/no questions give us numbers but don't tell us why people struggled. We don't know what specific navigation problems they hit or what would actually help. Open-ended questions or interviews would've been more useful.

Conclusion

The evaluation shows we've built something that works but isn't particularly usable yet. The 100% negative response on engagement and 70% failure rates on navigation and responsiveness make it clear that fixing the user experience must be the top priority.

That said, being honest about these problems and understanding what's wrong with our methodology shows we're thinking critically about evaluation rather than just chasing good numbers.

Critical reflection on design trade-offs

During development, we had to make some tough calls about what not to build. This section explains those decisions and why they were sensible scope management rather than us failing to deliver.

Trade-off 1: No Comment Moderation

Decision: Comments post immediately without curator approval or user flagging.

Why: Building a proper moderation system (flagging, queue interface, email notifications, spam detection) would take around 8-13 story points. We therefore prioritised core functionality instead.

Risk Management: We mitigated this by requiring login to comment, limiting comments to 2,000 characters, and allowing users to delete their own comments.

Trade-off 2: No Analytics Dashboard

Decision: We didn't build curator analytics showing most-viewed exhibits or quiz completion rates.

Why: Analytics would take around 10-13 story points and raises privacy concerns. Tracking which exhibits users view or which quizzes they fail conflicts with our data minimisation commitment.

Outcome: Curators can query the database directly for basic analytics. This avoids privacy issues while maintaining the option to add analytics later if needed.

Trade-off 3: Web Application Only

Decision: We built a responsive website, not native iOS/Android apps.

Why: Native apps would require learning new frameworks, implementing offline mode, app store accounts and maintaining separate codebases. All of this would have been 15+ story points for uncertain benefit.

Outcome: Having web application means no installation required, instant updates, and cross-platform support by default. All 20 evaluation participants successfully used the web application on their own devices.

Trade-off 4: Django Framework

Decision: We chose Django over alternatives like Flask.

Why: Django's built-in admin interface let curators manage exhibits without us building custom interfaces which in turn saves us around 13-21 story points.

Outcome: Django's admin proved essential for curator workflows.

Conclusion

These trade-offs demonstrate consistent prioritisation: we delivered complete, high-quality core features rather than attempting incomplete implementations of everything. We distinguished "must have" from "nice to have," respected our ethical commitments by avoiding analytics that would conflict with data minimisation and scoped realistically to our sprint capacity.

The evaluation revealed significant User experience issues due to our results having 70% navigation failure, 70% responsive design problems and 100% negative engagement responses. However, these are fixable improvements to an otherwise solid technical foundation. Had we attempted to build everything simultaneously, we would likely have delivered fragile software with broken core functionality. Our focused approach meant we built less, but what we built works correctly: 28 automated tests pass at 100%, curators can manage content efficiently and the core educational loop functions as designed.

Ethical and Legal Update

Summary

The Sprint 2 prototype still complies with everything we documented in Sprint 1. We haven't changed what data we collect, how we secure it, or how users interact with the system in any way that would need ethics approval or change our risk assessment.

Data Handling Confirmation

Our data collection practices remain minimal and unchanged:

Data Collected:

- Username
- Email Address
- Password
- Quiz Scores
- Comments

Data Not Collected:

- Sensitive personal data
- Location tracking
- Third-Party tracking cookies

User Safety Confirmation

Security and safety measures remain unchanged from Sprint 1:

- Passwords hashed using Django
- Role-based access control (visitors, curators, admins)
- Authentication required to post comments
- Comment body limited to 2,000 characters
- Users can delete their own accounts via profile page

Ethics Approval Decision - Reconfirmed

Our Sprint 1 conclusion that formal ethics approval is not required remains valid:

- Target audience: University students (18+) and professionals (no vulnerable groups)
- Minimal data collection: Only authentication credentials and quiz scores
- No sensitive data collected
- Voluntary participation in all interactions
- Low-risk educational purpose
- No third-party data sharing
- Transparent about system purpose

Conclusion

The final system meets the same ethical and legal standards we set in Sprint 1. The evaluation found major user experience problems, but those are design issues, not ethics issues. We're still committed to minimal data collection, voluntary participation, and transparency. The full ethical and legal considerations document is at 4_ethics_and_licensing/ethics_and_legal_considerations_final.pdf and it's still accurate for the current system.

Future Roadmap

Based on what the Sprint 2 evaluation told us, here are five improvements we'd tackle in a hypothetical Sprint 3. Each one includes what we'd build, why it matters, and how much work it'd be.

Improvement 1: Fix the Navigation (Critical)

What: Redesign the main navigation menu. Add a sticky header, clear icons and build a proper mobile menu.

Why: 70% of people couldn't find or use the navigation. That's catastrophic as if users can't find the menu, they can't get to any of the exhibits. Doesn't matter how good the content is if nobody can reach it.

How: Sticky header that doesn't disappear when you scroll, icons for main sections, proper menu for mobile.

Effort: 8 story points

Improvement 2: Fix Responsive Design (Critical)

What: Go through the CSS and fix the breakpoints so the layout works properly on phones, tablets, and desktops.

Why: 70% had layout problems when resizing their browser. Modern websites have to work on all screen sizes.

How: Test at different screen widths to find where it breaks, fix the layouts to be flexible instead of fixed-width, and check it all works in Chrome, Firefox, Safari, and Edge.

Effort: 5 story points

Improvement 3: Create a Design System (High Priority)

What: Standardise the fonts, colours, button styles, and spacing so everything looks consistent.

Why: 60% noticed inconsistent design. Different fonts and colours on different pages looks sloppy and unprofessional. Makes people doubt whether they can trust the content.

How: Pick standard heading sizes, choose a colour palette that we actually stick to, standardise how buttons and forms look, write it all down so the team knows what to use.

Effort: 8 story points

Improvement 4: Make It Less Boring (High Priority)

What: Add images, improve layouts, add some subtle animations, make the interface more visually interesting.

Why: 100% found it boring. Everyone. Not a single person thought it was engaging. Educational sites need to hold people's attention or they just leave.

How: Add decent images to exhibit cards, vary the layouts instead of using the same grid everywhere, add hover effects and smooth transitions, use bigger headings with better spacing, show summaries that expand when you click.

Important: We'd need to keep it accessible and have animations that can be turned off, proper alt text on images, colours that meet accessibility standards.

Effort: 13 story points

Improvement 5: Add Comment Moderation (Medium Priority)

What: Build a system where users can flag dodgy comments and curators can review and remove them.

Why: Right now comments go live immediately with no oversight. For a real production system, that's asking for trouble as the comments could have spam, hate speech or misinformation. We need moderation.

How: Add a "Flag as inappropriate" button on comments, build an admin page where curators can see flagged comments and approve or delete them, send email alerts when something gets flagged, automatically hide comments that get 3+ flags.

Effort: 8 story points

Prioritisation Summary

The five improvements are prioritised based on evaluation evidence strength, impact on core functionality and ethical/legal risk:

Critical (Weeks 1-3):

- Navigation Redesign (8 points) - 70% failure rate blocks content access
- Responsive Framework Fix (5 points) - 70% experience broken layouts

High (Weeks 4-6):

- Design System Implementation (8 points) - 60% report inconsistent design
- Interface Engagement Enhancements (13 points) - 100% found interface boring

Medium (Weeks 7-8):

- Comment Moderation System (8 points) **-** Prevents spam and harmful content

Total Estimated Effort: 42 story points (8 weeks of development)

We prioritised critical usability failures first followed by visual improvements that enhance engagement and finally safety features needed for production deployment. This approach ensures users can access and engage with content before adding community features.